

Spatiotemporal Taxi Demand Prediction with Novel Clustering and Demand Disaggregation Framework

Pura Nagaraja Uttunga

Department of Electronics and Communication
PES University, Bengaluru, India
Email: pes1202102729@pesu.pes.edu

Ritvik S Raiker

Department of Electronics and Communication
PES University, Bengaluru, India
Email: pes1202102454@pesu.pes.edu

Ravikumar Arjun

Department of Electronics and Communication
PES University, Bengaluru, India
Email: pes1202102446@pesu.pes.edu

Vamsi Krishna Tumuluru

Department of Electronics and Communication
PES University, Bengaluru, India
Email: vamsikrishnatumuluru@pesu.pes.edu

Abstract—Accurate taxi demand forecasting is essential for optimizing urban transportation services and resource allocation. However, spatiotemporal variations, external factors, and heterogeneous demand patterns make accurate prediction challenging. Unlike existing works, this study introduces a novel non-neighbor clustering approach to improve taxi demand prediction by grouping taxi zones based on demand correlation rather than geographic proximity. The proposed clustering method integrates Kernel PCA and K-Means algorithm. Different predictors such as Random Forest, XGBoost, AdaBoost, and LSTM are compared with and without the proposed clustering method. To improve the training performance of the predictors, a scenario reduction method is applied. Additionally, a demand disaggregation framework is proposed to break down cluster-level prediction to the taxi zone-level. Among the evaluated models, Random Forest achieved the best results for the taxi zone-level prediction with an accuracy of approximately 80%. We provide the comparison with benchmark work to highlight the performance of the proposed model.

Index Terms—Taxi Demand Prediction, Spatiotemporal Forecasting, Non-Neighbor Clustering, Demand Disaggregation, Machine Learning, Scenario Reduction, Random Forest.

I. INTRODUCTION

Indian taxi market share is expected to grow to USD 38.90 billion by 2029 at a CAGR of 13.55% [1]. Accurate taxi demand prediction is expected to play an important role in this growth. Taxi demand prediction is essential for effective urban transportation management, enabling improved service delivery and optimized resource allocation. One of the primary challenges in taxi demand prediction is the dynamic nature of spatiotemporal variations [2]. In other words, the number of pickups vary in both time (e.g., peak and off-peak hours) and across the geographic region (e.g., commercial regions and residential regions). Taxi demand prediction is typically carried out by dividing the dataset (e.g., number of pickups) of a larger geographic region into smaller datasets with respect to the different smaller taxi zones [3]–[5]. Zones can be visualized as squares of fixed areas overlaying the geographic region [6] or districts with corresponding boundaries [7]. The demand data in any zone is treated as a time series of pickups.

Historical time series of each zone is divided across the time, e.g., over 30-minute intervals [3], [8], 1-hour intervals [4], [9] or 2-hour intervals [5]. Then, the time series data is used to train the predictors. In some works, the time series data also includes additional attributes such as weekday or weekend patterns [6] or meteorological information [3]–[5], [8].

The performance of the predictor depends on the correlation in the time series used for training. Performance is usually better when the time series are aggregated across all the taxi zones and the predictor estimates the aggregate demand for the geographic region. Such an approach is referred to as *centralized prediction*. However, existing works using centralized prediction do not offer demand disaggregation methods to map the aggregated prediction back to the various taxi zones. Further, such an approach does not support creation of hotspots of taxi demand which could help the drivers. This issue can be addressed by predicting the taxi demand for groups of taxi zones. This approach is referred to as *distributed prediction*. Existing works have typically relied on proximity of the taxi zones when grouping them. However, the performance of the predictors could vary drastically across the clusters due to the arbitrariness in grouping the taxi zones.

Various models have been used for taxi demand prediction such as *historical average* (HA), *autoregressive integrated moving average* (ARIMA), *long-short-term memory* (LSTM), etc. Table I provides detailed comparison of the existing works related to taxi demand prediction. Among the existing works, only [6] performed a temporal analysis and concluded that 30-minute intervals yield the best performance in terms of *root mean square error* (RMSE). It was also observed that very few works ([3] and [10]) considered clustering methods that did not rely on proximity. However, it should be noted that none of the existing works discussed methods to map the prediction results of a cluster back to the taxi zones within the cluster. Motivated by these limitations, this paper proposes a novel distributed taxi demand prediction for the individual taxi zones. The proposed work consists of three stages: clustering of taxi zones, prediction of the aggregate taxi demand for

TABLE I
COMPARISON OF EXISTING WORKS

Paper	Prediction	Models	Clustering	Time Horizon	Evaluation Metrics
[3]	Distributed	MLRNN	Correlation and deterministic annealing	30 mins	MAE, RMSE, MAPE
[4]	Centralized, Distributed	RFM, RRM, CFM	Proximity based	1 hour	MSE, MAE, R ²
[5]	Centralized	pmlLSTM	Not used	2 hours	MAE, RMSE
[7]	Centralized	Stream Neural Network	Not used	30 mins	MAPE
[6]	Distributed	LSTM-MDN	Proximity based	5, 10, 20, 30, 60 mins	sMAPE, RMSE
[8]	Centralized	BERT	Not used	30 mins	RMSE, MAPE
[9]	Centralized	CCM	Not used	1 hour	R ² , MAE
[10]	Distributed	EWMA	DBSCAN	30 mins	Not applicable

each cluster and mapping of the predicted taxi demand back to the each taxi zone in the cluster. For the clustering, we propose a novel clustering scheme where the correlation values between the historic time series of different taxi zones is first constructed. Following this, the taxi zones are clustered by applying the Kernel *principal component analysis* (PCA) and k-means algorithm [11]. For predicting the aggregate demand of any cluster, we tested various models such as Random Forest, XGBoost, AdaBoost, and LSTM [11]. For disaggregating the predicted demand of a cluster back to the corresponding taxi zones, this work proposes using multi-output regressors (based on Random Forest, XGBoost, AdaBoost, LSTM plus Random Forest). The performance of the proposed work was analyzed in terms of RMSE, *mean absolute percentage error* (MAPE), *mean absolute error* (MAE) and R^2 . Comparisons with [3] and [6] showed significant improvements in the performance.

II. PROPOSED TAXI DEMAND PREDICTION SCHEME

This section explains the proposed distributed taxi demand prediction scheme to predict the demand at the taxi zone-level. Similar to the existing works, we assume that historic dataset related to taxi pickups is available across a large geographic region. It should be noted that historic datasets can be sparse across the geographic region and further contain duplicate entries. Therefore the dataset should be cleaned before use. The geographic region is then divided into a set of taxi zones, denoted by $\mathcal{L}$. Similar to [6], we consider one slot duration to be 30 minutes. Form the historic data related to taxi zone $l \in \mathcal{L}$, the number of pickups occurring over T slots is represented as a time series denoted $\mathbf{x}_l = \{x_{l,t}\}$, where $t \in \{1, \dots, T\}$. As the goal is to predict at the taxi zone-level in a large geographic region, the proposed scheme is implemented in three steps: a) clustering, b) predicting at cluster level and c) mapping the cluster-level prediction values to the taxi zone level. Next, we explain each of these steps.

A. Proposed clustering method

The proposed clustering method groups the taxi zones into clusters based on the similarity between the taxi zones rather than the proximity between them. Here, the similarity between any pair of taxi zones $l, l' \in \mathcal{L}$ is expressed using the normalized cross-correlation, denoted by $r(\mathbf{x}_l, \mathbf{x}_{l'})$. The normalized cross-correlation is defined below where μ_l and

$\mu_{l'}$ represent the time averages of the time series $\mathbf{x}_l$ and $\mathbf{x}_{l'}$, respectively.

$$r(\mathbf{x}_l, \mathbf{x}_{l'}) = \frac{\sum_{t=1}^T (x_{l,t} - \mu_l)(x_{l',t} - \mu_{l'})}{\sqrt{\sum_{t=1}^T (x_{l,t} - \mu_l)^2 \sum_{t=1}^T (x_{l',t} - \mu_{l'})^2}} \quad (1)$$

It should be noted that $r(\mathbf{x}_l, \mathbf{x}_{l'}) \in [-1, 1]$. We convert these $L \times L$ correlation values into a dissimilarity matrix $\mathbf{D}$ where $L = |\mathcal{L}|$. The (l, l') -th element of the dissimilarity matrix is given by $D(\mathbf{x}_l, \mathbf{x}_{l'}) = 1 - r(\mathbf{x}_l, \mathbf{x}_{l'})$. The dissimilarity provides a measure of distance between two time series in the feature space. $D(\mathbf{x}_l, \mathbf{x}_{l'}) \in [0, 2]$ where a value of 0 implies a perfect positive correlation and 2 implies a perfect negative correlation.

Given a large L , we create the clusters using the dissimilarity matrix $\mathbf{D}$ as follows. First, the Gaussian kernel is used to convert the $\mathbf{D}$ into an $L \times L$ kernel matrix $\mathbf{G}$. The (l, l') -th element of $\mathbf{G}$ is expressed as $\exp(-(\mathbf{D}_l - \mathbf{D}_{l'})^2/L)$ where $\mathbf{D}_l$ and $\mathbf{D}_{l'}$ represent the l -th and l' -th rows of $\mathbf{D}$ respectively. Next, we determine the centers of the K clusters using principal component analysis [12]. Given the centers $\mathbf{c}_k$, where $k \in \{1, \dots, K\}$, the taxi zones are matched to the centers using K-means algorithm. The cluster k corresponding to taxi zone $l \in \mathcal{L}$ is given by $k = \arg \min_k \|\mathbf{G}_l - \mathbf{c}_k\|$ where $\mathbf{G}_l$ represents the l -th row of the kernel matrix $\mathbf{G}$.

B. Aggregation and prediction at cluster-level

Let $\mathcal{C}_k$ denote the set of taxi zones that belong to the cluster $k \in \{1, \dots, K\}$ obtained using the proposed clustering method. The time series of the pickups associated with the taxi zones are aggregated to form the new time series per cluster k . The aggregated time series is denoted by $\mathbf{y}_k = \{y_{k,t}\}$, where $y_{k,t} = \sum_{j \in \mathcal{C}_k} x_{j,t}$. The input and the desired output of the predictor for cluster k are denoted by $[y_{k,t-1}, \dots, y_{k,t-w}]$ and $y_{k,t}$ respectively. Here, w represents the order of the predictor. Due to the large dynamic range of the pickups occurring in the aggregated demand at the cluster-level, we normalize the input and the desired outputs prior to training. The output of the predictor is re-scaled and rounded off after training. Using the inputs and outputs, various predictors such as Random Forest, XGBoost, AdaBoost, and LSTM [11] are trained using supervised learning. From our analysis, the Random Forest algorithm provided the best performance.

Therefore, we discuss the next section based on cluster-level predictions performed by the Random Forest algorithm.

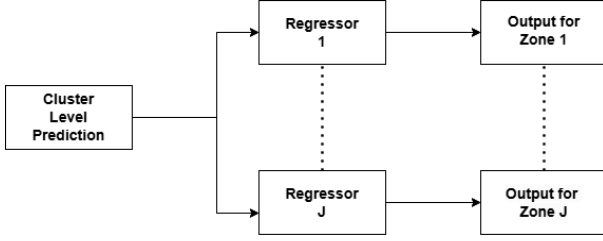


Fig. 1. Schematic of the proposed taxi-zone level prediction model

C. Obtaining taxi zone-level predictions

The last step of the proposed taxi demand prediction scheme involves disaggregation of the cluster-level predictions into taxi zone-level predictions. The aggregate taxi demand predicted for the cluster k in time slot t is denoted by $\hat{y}_{k,t}$. This aggregate taxi demand should be disaggregated across all the taxi zones $j \in \mathcal{C}_k$. In order to accomplish this, we introduce a multi-regressor model as shown in Fig. 1. The multi-regressor model corresponding to cluster k consists of J regressors where $J = |\mathcal{C}_k|$. We experimented with different models for the regressors. The input to all the regressors is given by $[\hat{y}_{k,t}, \dots, \hat{y}_{k,t-v}]$ where v is the order of the regressors. The output of the j -th regressor is given by $z_{\mathcal{C}_k(j),t}$. Due to the large dynamic range of the pickups occurring in the taxi zone level demand, we normalize the input and the desired outputs prior to training. The output of the regressors are re-scaled and rounded off after training. The regressors are simultaneously trained by minimizing the loss function $\mathbb{L}(\hat{y}_{k,t})$ given below, where $x_{\mathcal{C}_k(j),t}$ is the desired predicted value of the j -th taxi zone in cluster $\mathcal{C}_k$ in time slot t based on the corresponding historic dataset. Here, $\mathbb{E}$ represents the expectation taken over the time and across all taxi zones in the cluster k .

$$\mathbb{L}(\hat{y}_{k,t}) = \mathbb{E}[(z_{\mathcal{C}_k(j),t} - x_{\mathcal{C}_k(j),t})^2] \quad (2)$$

III. RESULTS AND ANALYSIS

This section presents the results and detailed analysis of the cluster-level prediction, taxi zone-level prediction and the comparative study.

A. Dataset and Evaluation criteria

This study used the NYC-Taxi-2018 dataset [13] from January to December 2018. Each record in the dataset includes *Taxi Zone Numbers* corresponding to 263 taxi zones, *timestamps* for each pick-up or drop-off event. The pickups in the dataset were converted into time series with a slot duration of 30 minutes similar to [6]. The dataset was split as 80% for training and 20% for testing. The performance is evaluated in

terms of MAE, RMSE, MAPE and R^2 . The expressions for the performance evaluation are given below.

$$MAE_i = \frac{1}{T} \sum_{t=1}^T |r_{i,t} - \hat{r}_{i,t}| \quad (3)$$

$$RMSE_i = \sqrt{\frac{1}{T} \sum_{t=1}^T (r_{i,t} - \hat{r}_{i,t})^2} \quad (4)$$

$$MAPE_i = \frac{1}{T} \sum_{t=1}^T \frac{|r_{i,t} - \hat{r}_{i,t}|}{r_{i,t}} \times 100\% \quad (5)$$

$$R_i^2 = 1 - \frac{\sum_{t=1}^T (r_{i,t} - \hat{r}_{i,t})^2}{\sum_{t=1}^T (r_{i,t} - \bar{r}_i)^2} \quad (6)$$

Here, $r_{i,t}$ is the observed demand, $\hat{r}_{i,t}$ is the predicted demand and $\bar{r}_i$ is the mean observed demand of each taxi zone i . For the plots, the above metrics are averaged across the 57 taxi zones.

B. Cluster-level Performance

Fig.2 shows the clustering of the original 263 taxi zones of New York based on the 2018 dataset. It can be observed that 24 clusters were formed using the proposed clustering scheme and the taxi zones are not adjacent to one another. This helps in mitigating the sparsity of demand at certain taxi zones. For the sake of comparing with [3], only the taxi zones having mean number of pick-ups of at least 10 per slot were retained. This resulted in 57 taxi zones similar to [3].

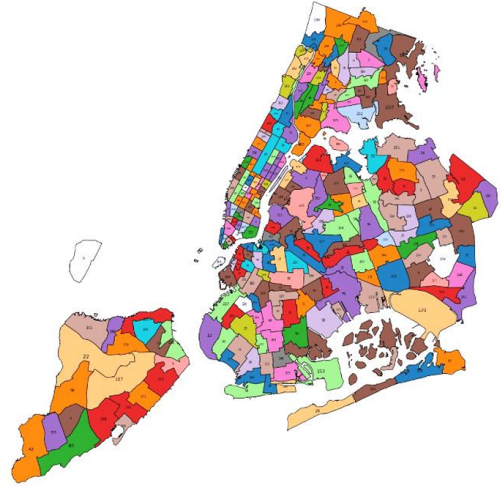


Fig. 2. Plot showing all Clusters with unique colors.

The proposed clustering scheme produced 8 clusters out of the 57 taxi zones. Figure 3 compares the actual taxi demand of Cluster 1 (solid blue curve) and the predicted taxi demand of Cluster 1 (dashed red curve). For this comparison, Random Forest method was used for the cluster-level prediction. It can be seen that the two curves are closely aligned with an $RMSE$ of 35.06. Similar trends were observed in other the clusters ($RMSE$ was in the range of 30 – 45).

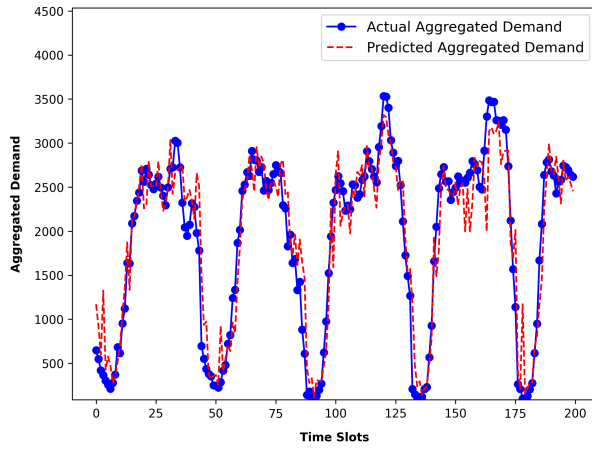


Fig. 3. Predicted demand versus actual demand in cluster 1.

Figure 4 evaluates the *RF* algorithm in terms of the RMSE (left y-axis) and the computation time (right y-axis) by varying the order of prediction (i.e., the number of past samples per input sample). It can be observed that the RMSE decreased significantly until the order of prediction v reached 12 slots, but started to increase thereafter due to a lack of correlation and redundancy in the samples. The convergence time decreased linearly with the order of the prediction. Due to the lower RMSE, we consider 12 as the best order of prediction.

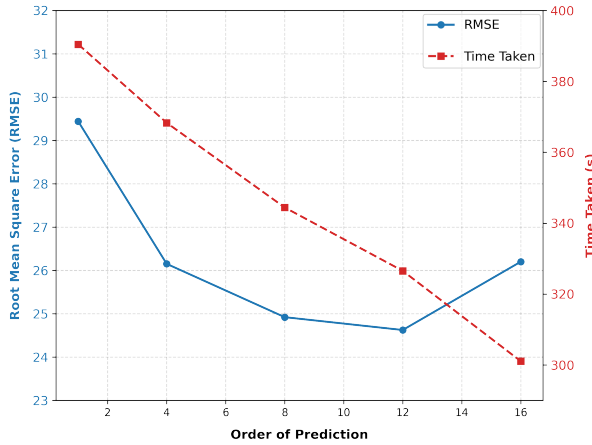


Fig. 4. RMSE and convergence time (in seconds) versus order of prediction.

C. Taxi zone-level performance

This section presents the taxi zone-level performance after the multi-output regressor disaggregated the cluster level predicted values. Different models/algorithms such as Random Forest, XGBoost, AdaBoost, and an LSTM+RF hybrid were used for constructing the multi-regressors. The analysis was carried out for every cluster and the results were averaged across all the taxi zones.

Fig.5 shows the RMSE achieved for different types of multi-output regressor. It can be observed that the Random Forest algorithm and XGBoost model achieved RMSE values of

25.95 and 26.42 respectively. Whereas, the AdaBoost model and LSTM+RF hybrid model yielded higher RMSE values of 31.56 and 30.47 respectively. This suggests that Random Forest algorithm produced more precise predictions for any taxi zone compared to the other models. In other words, the average variation between the residuals (difference between the actual value and predicted values) was less under the Random Forest algorithm.

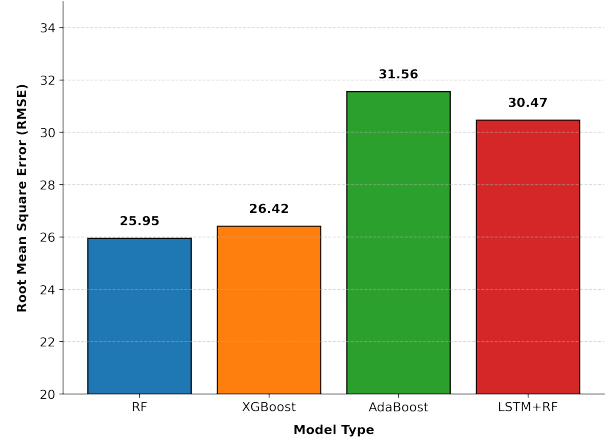


Fig. 5. Comparison of RMSE for different multi-regressors.

Fig.6 shows the comparison of the MAE observed under different multi-output regressor. MAE provide additional insights into the prediction accuracy of the models. MAE represents the mean absolute difference between predicted and actual values. It can be seen that the RF and XGBoost models exhibited lower MAE values of 18.77 and 19.17, respectively. Whereas the AdaBoost and LSTM+RF exhibited higher MAE values of 23.99 and 23.28 respectively. These findings reaffirm that RF and XGBoost generate more stable predictions with smaller average deviations.

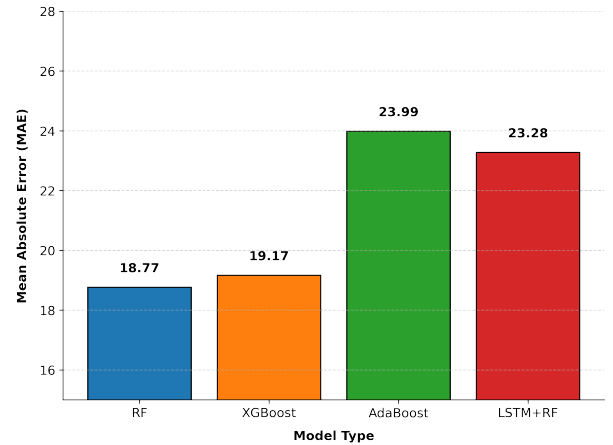


Fig. 6. Comparison of model performance across MAE.

Fig.7 shows the comparison of the R^2 observed under different multi-output regressor. It can be observed that the RF

and XGBoost models achieve a higher R^2 score of 0.83 and 0.73 (i.e., 83% and 73%) respectively signifying low variance in the predicted demand and better fitting to the actual demand. In contrast, the AdaBoost and LSTM-RF models exhibited lower R^2 values of 0.59 and 0.69 respectively.

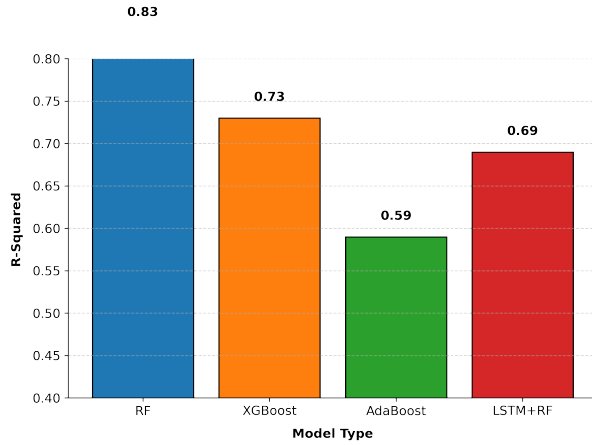


Fig. 7. Comparison of model performance across R^2 .

Fig.8 shows the comparison of the MAPE observed under different multi-output regressor. It can be observed that the RF and XGBoost models yielded the lowest MAPE values of 35.88% and 35.62%, respectively. In contrast, the AdaBoost and LSTM-RF models had substantially higher MAPE values of 55.30% and 52.33% respectively, suggesting that these models struggle with precision, particularly when applied to datasets with large dynamic range.

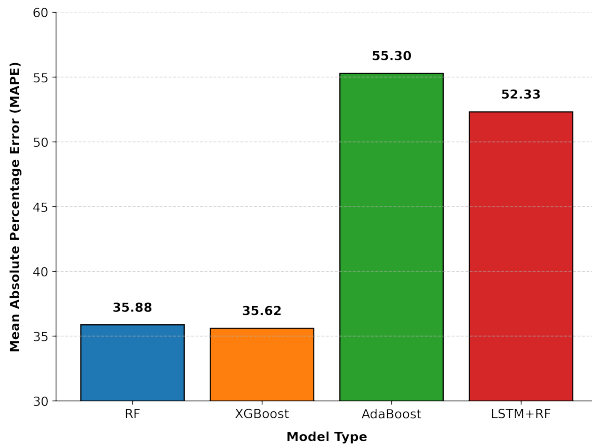


Fig. 8. Comparison of model performance across MAPE.

Owing to the better performance across all metrics, we analyze the taxi zone-level prediction for other results using the the RF algorithm. Fig. 9 shows the comparison of the taxi demand predicted by the proposed multi-output regressor and the actual demand for a dense taxi-zone (index 107) from cluster 1. Fig. 10 shows the comparison of the taxi demand predicted by the proposed multi-output regressor and the actual demand for a sparse taxi-zone (index 125) from cluster 1. The

close alignment of the graphs in both figures demonstrates the model's robustness in forecasting location-specific demand trends.

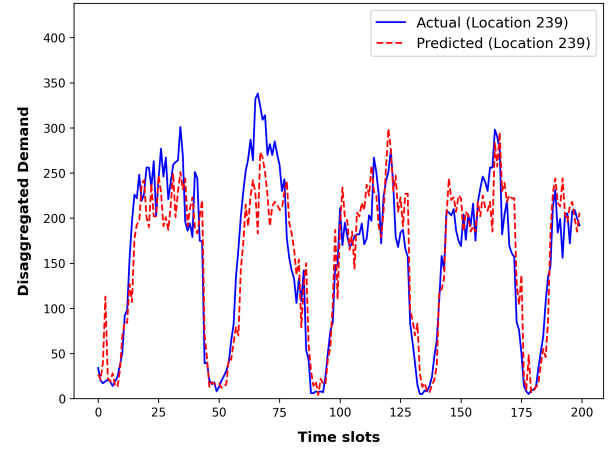


Fig. 9. Prediction performance at a dense taxi zone 239.

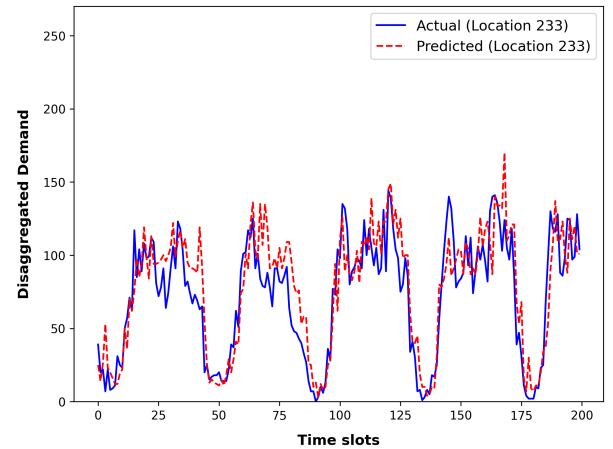


Fig. 10. Prediction performance at a sparse taxi zone 233.

Table II highlights the impact of proposed clustering algorithms on the performance of the taxi zone-level prediction. The RMSE averaged across all taxi zones showed significant improvement when both the kernel PCA and the K-means algorithm were applied as proposed. Further, the convergence time of the Random Forest algorithm training the proposed multi-output regressor model also significantly decreased when both the kernel PCA and K-means algorithm were applied as proposed. This could be because the kernel PCA determined the optimal cluster centers and the K-means identified the best taxi zones for the clusters. It should be noted that the proposed algorithm was influenced by the proximity of the time series of the taxi-zones (i.e., cross correlation) rather than the physical proximity.

Finally, we compared the prediction performance of the proposed work with [3]. It should be noted that clustering was performed in [3] in two steps, first the Pearson correlation

TABLE II
TABLE SHOWING THE RESULTS WITH GAUSSIAN KERNELS
AND SCENARIO REDUCTION

Kernel PCA	K-means	RMSE	Training Time(in seconds)
No	No	28.45	759.67
No	Yes	26.16	425.51
Yes	No	28.23	1618.72
Yes	Yes	24.50	398.68

coefficients were calculated for any pair of taxi zones and a deterministic annealing algorithm was applied. Thereafter, the predictions of the taxi demand at the cluster-level were carried out using the MLRNN model. The 57 taxi zones were grouped into 4 clusters in [3]. The RMSE for cluster-level prediction reported in [3] was 23.08. For the same number of clusters, the proposed method produced an RMSE for cluster-level prediction as 28.97. However, none of the existing works, provided a method for disaggregation of the cluster-level demand into taxi zone-level demand. Therefore, a comparison of the taxi zone-level demand prediction cannot be provided. However, the average RMSE for predicting the demand at the taxi zone-level under the proposed work is given in Table III. The results demonstrate that as the number of clusters increases, the performance of the taxi zone-level prediction improves.

TABLE III
TAXI-ZONAL LEVEL PREDICTION ERROR METRICS ACROSS
CLUSTERS

No. of Clusters	RMSE	MAE	R-squared	MAPE
C-1	29.56	21.06	0.69	42.02
C-2	26.78	19.27	0.73	36.26
C-3	26.89	19.47	0.72	38.01
C-4	26.50	19.03	0.74	36.26
C-5	25.82	18.69	0.74	35.41
C-6	25.83	18.69	0.74	35.12
C-7	26.14	18.87	0.75	34.89
C-8	25.38	18.41	0.76	34.75
C-9	25.08	18.18	0.76	34.39
C-10	24.17	17.57	0.76	33.95

IV. CONCLUSION

This paper proposed a novel framework for spatiotemporal taxi demand forecasting, leveraging non-neighbor clustering and an innovative disaggregation model. The methodology integrated the Kernel PCA and K-means algorithm to perform clustering of the taxi zones. The importance of combining both methods for clustering was demonstrated in the results. The proposed disaggregation model, referred to as the multi-output regressors ensured that the cluster level prediction values were mapped down to individual taxi zone-level. Extensive experiments were conducted on the 2018 New York Taxi dataset. Upon analyzing various multi-output regressors models, we conclude that the best results were obtained when both the cluster-level predictors and the multi-output regressors used the Random Forest algorithm. The performance was evaluated using various metrics such as RMSE, MAE, R^2 and MAPE.

As the number of clusters were increased, the prediction performance at the taxi zone-level improved significantly. In the future work, we aim to further improve the performance of the taxi zone-level prediction which will enable the development of accurate forecasts of the demand hotspots to the drivers.

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